

EYA AZZABI

AI & Data Engineer | Final-Year Engineering Student

Ben Arous, Tunisia | +216 93 240 604 | eyaazzabi04@gmail.com | linkedin.com/in/eya-azzabi | github.com/EyaAzzabi | eyaazzabi.github.io

Final-year internship from Jan 2027 | Relocation: Europe | Canada | Worldwide

PROFESSIONAL SUMMARY

Final-year Data Science engineering student at ESPRIT, completing a Master's in Data Science in parallel, having worked as an AI & Backend Developer. **1st place nationally** at the ESPRIT / Ministry of Finance "AI & Public Finances" hackathon (Sep 2026). I build **LLM agent systems, privacy-preserving ML pipelines and computer-vision models** for production.

PROFESSIONAL EXPERIENCE

AI & Backend Developer, Mafrah (formerly WebFullContact), Remote **Sep 2025 – Sep 2026**

- Parsing CVs meant sending personal data to a cloud LLM, creating GDPR exposure.** Engineered a **pseudonymisation-before-inference pipeline** (Presidio + spaCy NER → GPT-4o → local re-identification), so the model never receives candidate identity and **GDPR Art. 5(1)(c)** holds by construction.
- Keyword search could not match consultants to roles reliably.** Implemented **semantic matching** with vector embeddings in **PostgreSQL/pgvector**, replacing exact-term lookup with retrieval over meaning.
- Agents lost context between turns and services could not scale independently.** Built a **custom agent orchestration layer** with memory and deployed **containerised FastAPI microservices** behind a **NestJS** backend, serving production users throughout the contract.

AI Consultant Intern, Talan Summer Camp, Talan Tunisie **2025**

- Sales teams lost hours triaging inbound leads by hand.** Built **task-oriented LLM agents** for lead prioritisation, forecasting and automated outreach inside **Connect'IS**, delivering a working end-to-end platform with a cross-functional team within the two-month camp.

PROJECTS

Jiser: AI platform for tax dispute resolution, 1st place National Hackathon **Sep 2026**

Python, FastAPI, React, XGBoost, hybrid RAG, Supabase, n8n, Docker; github.com/EyaAzzabi/Jiser

- 1st place nationally** (30 teams) at ESPRIT / Ministry of Economy & Finances' "AI & Public Finances" hackathon. Three AI systems (rule engine, **XGBoost risk model at AUC 0.84**, hybrid RAG over 305 tax articles) help resolve fiscal disputes before the 45 day deadline; prioritising the top 20% of cases by model score recovered **107.3M DT vs. 28.5M DT** for age based triage. Team of 4.

X-Ray: Detecting silent failures in AI agent conversations, 3rd place GoMyCode Hackathon **Sep 2026**

Python, FastAPI, React, TF-IDF/KMeans, LLM-as-judge; github.com/EyaAzzabi/SupplyzPro-Smart-Operations-Award

- 3rd place, GOMYCODE x NVIDIA Real-World AI Impact Award.** Detects agent failures that throw no error, clustering them by root cause: **100% precision/recall** on synthetic cases, validated at **32% recall** on 78 real transcripts. Team of 5.

JobReady: Master's thesis **2025 – 2026**

Python, GPT-4o, pgvector, Presidio, spaCy, FastAPI, NestJS, Angular, Docker

- CV data is unstructured, and the EU AI Act classifies employment AI as high-risk.** Designed a three-phase system spanning information extraction, semantic retrieval and LLM-driven interview simulation: **96.3% PII-detection recall** with **0% leakage** to the cloud model and **NDCG@5 = 0.948** on matching, on a **running multi-tenant platform** GDPR-compliant by construction. Defence autumn 2026.

QoS Buddy: Multi-agent telecom network diagnostics **2026**

Python, scikit-learn, pandas, JavaScript; qos-buddy.vercel.app

- Operators detect SLA breaches only after they occur.** Developed **four autonomous agents** for latency prediction, anomaly detection, congestion analysis and data enrichment, reaching **$R^2 = 0.941$** on latency forecasting. ESPRIT integrated project, team of six.

PCA of Tunisian Temperatures: Bachelor's thesis, extended and open-sourced **2024**

Python, NumPy, Matplotlib; github.com/EyaAzzabi/pca-tunisia-temperatures

- Twelve monthly variables were highly correlated, with no geographic labels supplied.** Applied normed **PCA** across 16 stations, reducing them to **two components explaining 98.4%** of variance and recovering the coastal-inland divide unsupervised.

TECHNICAL SKILLS

Languages	Python, SQL, R, Java, TypeScript, JavaScript
Machine Learning	PyTorch, scikit-learn, XGBoost, TensorFlow, YOLO, CLIP, OpenCV, CNNs, Transformers, LoRA fine-tuning, feature engineering, model evaluation
NLP & LLMs	LangChain, RAG, prompt engineering, Hugging Face, spaCy, NLTK, TF-IDF, embeddings, pgvector, FAISS, vector databases
Data Engineering	pandas, NumPy, Apache Spark, Hadoop, ETL pipelines, PostgreSQL, MySQL, MongoDB, Elasticsearch, Snowflake
MLOps & Cloud	AWS, Docker, Kubernetes, CI/CD, GitHub Actions, model deployment, REST APIs, FastAPI, NestJS, microservices, Git, Linux, Agile/Scrum
Statistics	PCA, inferential statistics, predictive modelling, Power BI

EDUCATION

Engineering Degree in Data Science, ESPRIT, Tunis, Tunisia **Sep 2025 – Oct 2027**

Master's Degree in Data Science, University of Carthage, Bizerte, Tunisia **Sep 2024 – 2026**

Ranked **1st of 25** (top 4%), average **14.75/20**; taken concurrently with the ESPRIT engineering degree, self-funded.

BSc Mathematics and Computer Science, University of Carthage, Bizerte, Tunisia **2021 – 2024**

Ranked **2nd of 20** (top 10%).

LEADERSHIP & COMMUNITY

Speaker, WYSE (Women & Youth in STEM Employment) and Radio Amal, on women in technology; Soliya Connect **2025 – 2026**

Program alumna

Project Developer, Enactus FSB; built SenseMate, an assistive AI device for blind users using real-time YOLO detection. **2023 – 2025**

Won the **Early Stage award**, Enactus National Exposition 2024

Co-founder, Association Youth Clubs; founded the first youth club in Morneg, Ben Arous **2019 – 2021**

CERTIFICATIONS & LANGUAGES

Applications of AI for Anomaly Detection, <i>NVIDIA</i>	Apr 2026	Developing Applications with LangChain, <i>DataCamp</i>	Feb 2025
Fundamentals of Deep Learning, <i>NVIDIA</i>	Jan 2026	Large Language Model Concepts, <i>DataCamp</i>	Feb 2025
AWS Academy Data Engineering, Graduate, AWS	May 2025	Getting Started with AI on Jetson Nano, <i>NVIDIA</i>	Jan 2025
Languages: Arabic (native) French (advanced, C1) English (advanced, C1) German (basic, A2)			